

Services / Technologies Required

Azure Reserved Instances and Savings Plan Analysis

Cloud Services

Service / Technology	Purpose
Microsoft Azure	Cloud platform and source of usage/billing data
Azure Cost Management	Cost analysis, usage views and pricing information
Azure Usage / Billing Data	Historical data used for analysis

Programming & Data Analysis

Service / Technology	Purpose
Python	Main programming language
Pandas	Data cleaning, processing and analysis
NumPy	Numerical calculations
Jupyter / VS Code	Development and analysis environment
Matplotlib	Charts and visualizations

Storage & Reporting

Service / Technology	Purpose
PostgreSQL	Data storage (if implemented)
Excel	Analysis and reporting
Power BI	Dashboard and visualization

Analysis Techniques

Service / Technology	Purpose
Descriptive statistics	Summarize usage patterns
Moving averages	Analyse usage trends
Usage variability	Measure workload stability
Cost-benefit analysis	Compare pricing options
Utilization analysis	Measure commitment usage
Threshold-based recommendation	Support final pricing recommendation

Note: PostgreSQL is marked 'if implemented' in the source PPT.